Iowa State University
Electrical And Computer Engineering

EE 230

LAB 3 REPORT

09/24/2025

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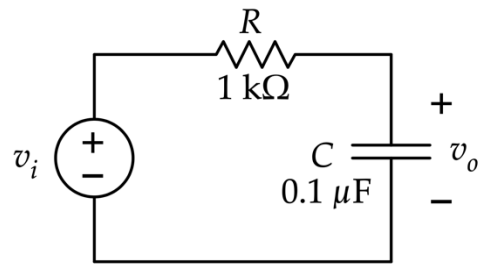
Introduction

This lab will require us to build, measure, and plot the circuits, results, and transfer functions of first order filter circuits. We will be using things like the loopinator, LMC660, capacitors, inductors, MATLAB, Excel, and the oscilloscope to help us carry out this lab.

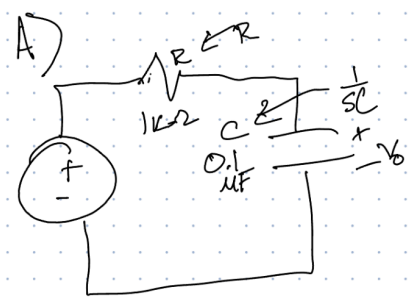
A RC Low-Pass Filter

Give the calculated transfer function and expected corner frequency for the circuit. Include relevant steps used to derive the transfer function. Also provide plots containing both measured and calculated frequency response.



If your measured values don't match the calculated values, explain why you think this is.

A)



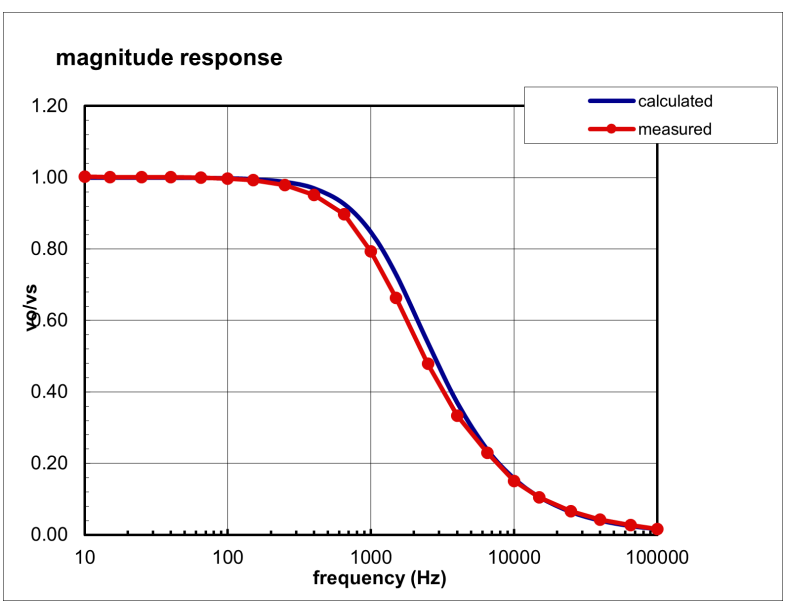
$$\frac{V_o}{V_i} = \frac{Z_C}{R+Z_C} = \frac{\frac{1}{sC}}{R+\frac{1}{sC}} = \frac{1}{1+sRC}$$

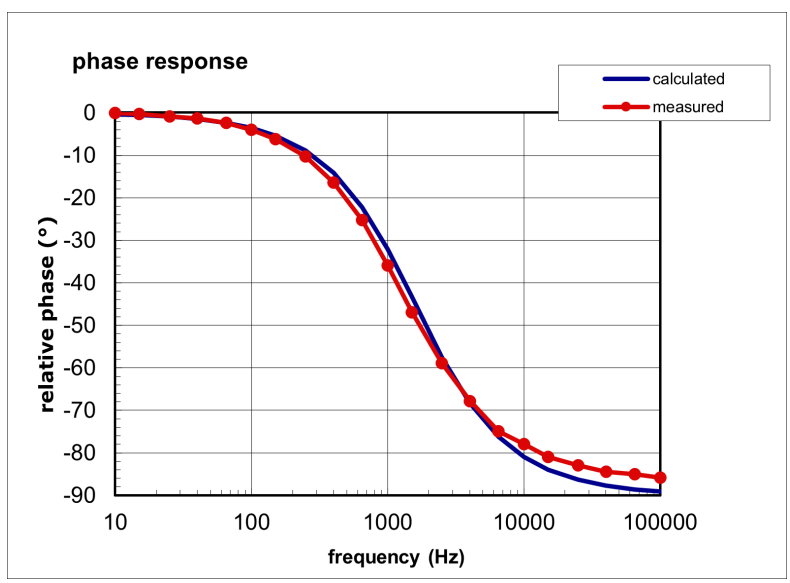
$$\omega_c = \frac{1}{RC}$$

$$f_c = \frac{\omega_c}{2\pi} = \frac{1}{2\pi RC}$$

$$|H(j\omega)| = \frac{1}{\sqrt{1+(\omega RC)^2}}$$

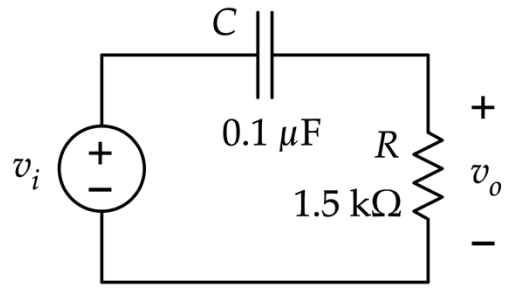
$$\tan^{-1}(\omega RC)$$

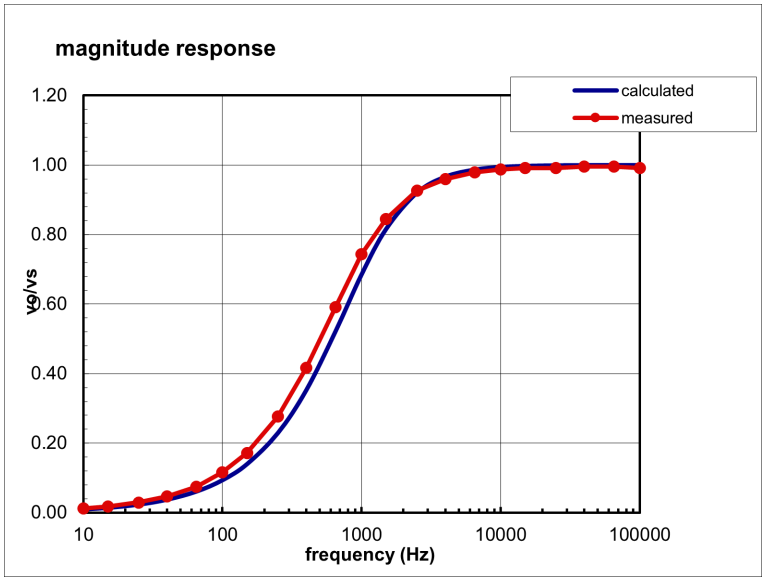


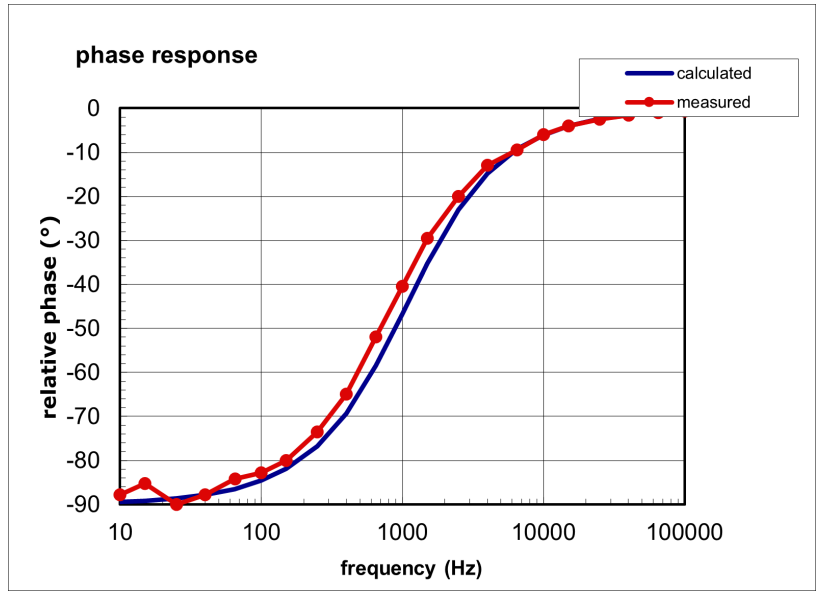


B RC High-Pass Filter

Give the transfer function and expected corner frequency for the circuit. Include relevant steps used to derive the transfer function. Also provide plots containing both measured and calculated frequency response.

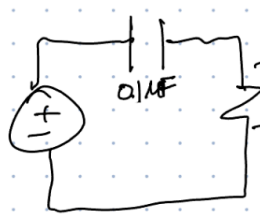






If your measured values don't match the calculated values, explain why you think this is

B)



$$\frac{v_o}{v_i} = \frac{R}{\frac{1}{sC}+R} = \frac{sRC}{1+sRC} = \frac{j\omega RC}{1+j\omega RC}$$

$$H(j\omega) = \left(\frac{j\omega RC}{1+j\omega RC}\right)\left(\frac{1-j\omega RC}{1+j\omega RC}\right) = \frac{(\omega RC)^2 + j\omega RC}{1+(\omega RC)^2}$$

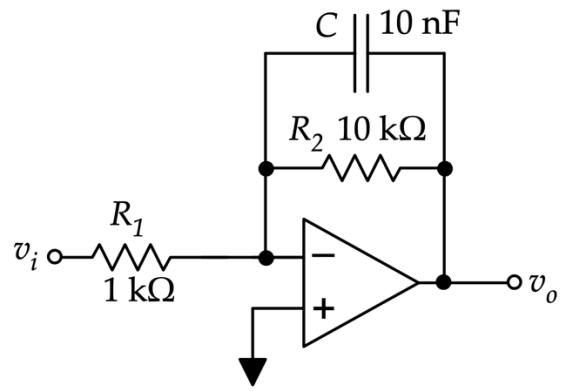
Real: $\frac{(\omega RC)^2}{1+(\omega RC)^2}$, imag: $\frac{\omega RC}{1+(\omega RC)^2}$

$$|H| = \sqrt{Real^2 + imag^2} = \boxed{\frac{\omega RC}{\sqrt{1+(\omega RC)^2}}}$$

$$\angle H = \tan^{-1}\left(\frac{imag}{Real}\right) = \tan^{-1}\left(\frac{1}{\omega RC}\right) = \boxed{90^\circ - \tan^{-1}(\omega RC)}$$

C Active Low-Pass Filter

Give the transfer function of the circuit. Include relevant steps used to derive the transfer function.



In the table below provide your expected pass-band gain, in units of V/V, and corner frequency.

Parameter	Calculated Value
Pass-Band Gain	-10 V/V
Corner Frequency (f_c)	1.592 kHz

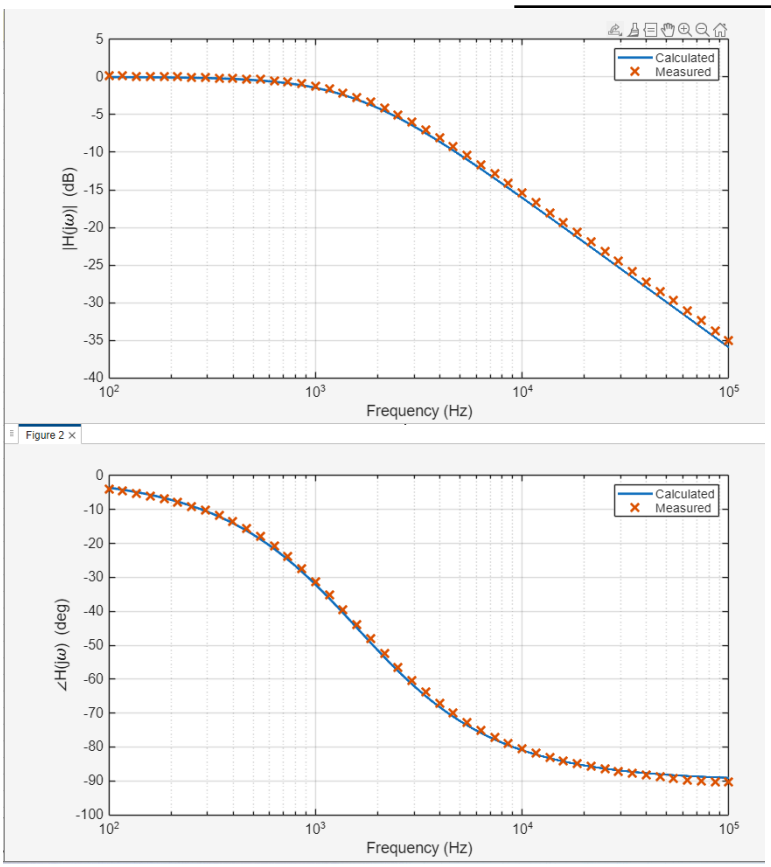
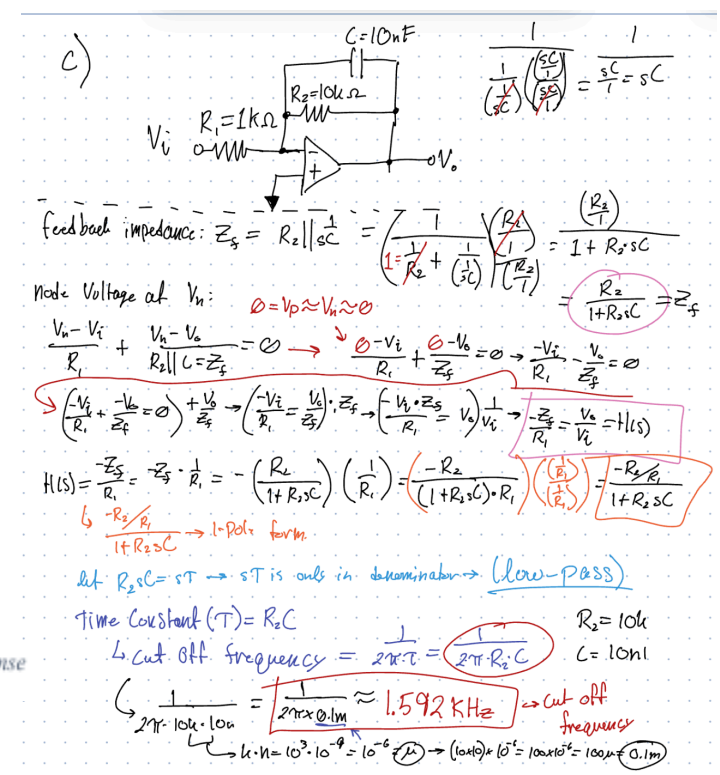
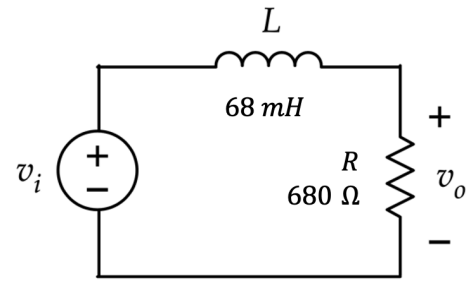


Figure 5: Bode Plot (Magnitude and Phase) of Measured Vs. Calculated Frequency Response

Comment on the above plot.
The plot to the side almost perfectly matches what should have been. We used the script in provided to us on Canvas to generate a plot from our data and also to generate the function it nearly matches.



D RL Low-Pass Filter



Give the transfer function of the circuit. Include relevant steps used to derive the transfer function.

In the table below, fill in your calculations and measurements from the lab procedure.

Measurement	Value
Ideal Low-Frequency Magnitude	$\lvert H(\sim 0)\rvert = 1\ V/V$
Ideal Cutoff Frequency	$f_c = 1.59\ kHz$
Measured Low-Frequency Magnitude	$\lvert H(\sim 0)\rvert = .75\ V/V$
Measured Cutoff Frequency	$f_c = 2.12\ kHz$
Measured Inductance ($f\ =\ $)	$L = 66.76mH$
Measured DCR	$R_L = 178.7Ohms$
Non-Ideal Low-Frequency Magnitude	$\lvert H(\sim 0)\rvert = 1$ V/V
Non-Ideal Cutoff Frequency	$f_c = 1.5$kHz

The values we obtained for our measured/calculated values of our ideal and non-ideal conditions were similar, but not quite so much our measured values for our cut frequency. We are not quite sure what may have caused this issue other than a faulty measurement. In this lab we have found that the bus strips on our breadboards make our measurements very inaccurate. We think there is some kind of resistance within the breadboard especially so on those bus strips. We think that this may have been the cause of it and we either forgot to account for it or we didn't know about it.

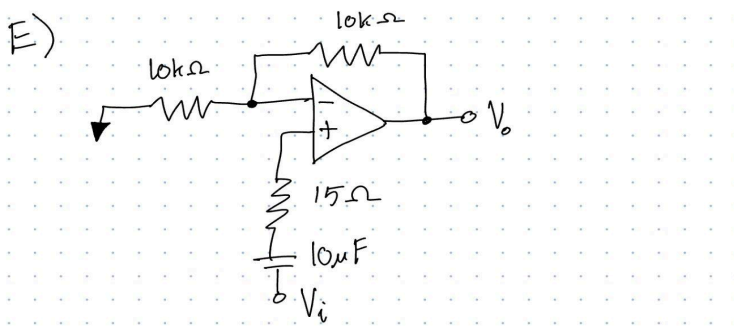
$V_o = |H(\sim 0)| / \sqrt{2}$ at the circuit's cutoff frequency because the circuit is a 1st order filter low-pass filter. This means that the cutoff frequency is equal to the total resistance, which makes the equation equivalent to its DC value.

D) $H(s) = \frac{V_o}{V_i} = \frac{R}{R+Z_L} \rightarrow \frac{1}{1+\frac{sL}{R}} \rightarrow \frac{1}{1+\frac{j}{\omega_c}} \rightarrow \omega_c = \frac{R}{L}$

$|H(j\omega)| = \frac{R}{\sqrt{R^2+(\omega L)^2}}$ $\quad \angle H(j\omega) = -\tan^{-1}\left(\frac{\omega L}{R}\right)$

$H(0) = 1\ V/V$ $\quad f_c = \frac{\omega_c}{2\pi} = \frac{R}{2\pi L} = \frac{680}{2\pi \cdot .068} = 1.6KHz$

E Active High-Pass Filter Design



$Z_c + 15\Omega = R_i$

$R_f = 10k\Omega$ $\quad$ Gain $= 1 + \frac{10k\Omega}{10k\Omega} = 2$

$f_c = \frac{1}{2\pi \cdot R \cdot C} = \frac{1}{2\pi \cdot 15 \cdot 10\mu} = \frac{1}{300\pi \cdot (10^{-6})} = \frac{1}{3\pi(10^{-4})}$

$\frac{1}{3\pi(10^{-4})} = 1.061\ Hz = 1.061\ kHz \approx 1\ kHz$

Briefly describe your design process and include your design calculations. Insert a sketch of your circuit with labeled component values below

I got this idea for the circuit from my lab partner but the values (bottom resistor value only) didn't work out. I plugged and checked different values for the resistor going into the positive op amp terminal until I got close to 1kHz for my cut of frequency.

Figure 6: Sketch of High-Pass Filter Design

Provide your "quick-and-dirty" measurements:

Description	Value
Cutoff Frequency	$f_c = 1.061$ kHz
Pass-Band Gain	$G_{pass} = 2$ v/v

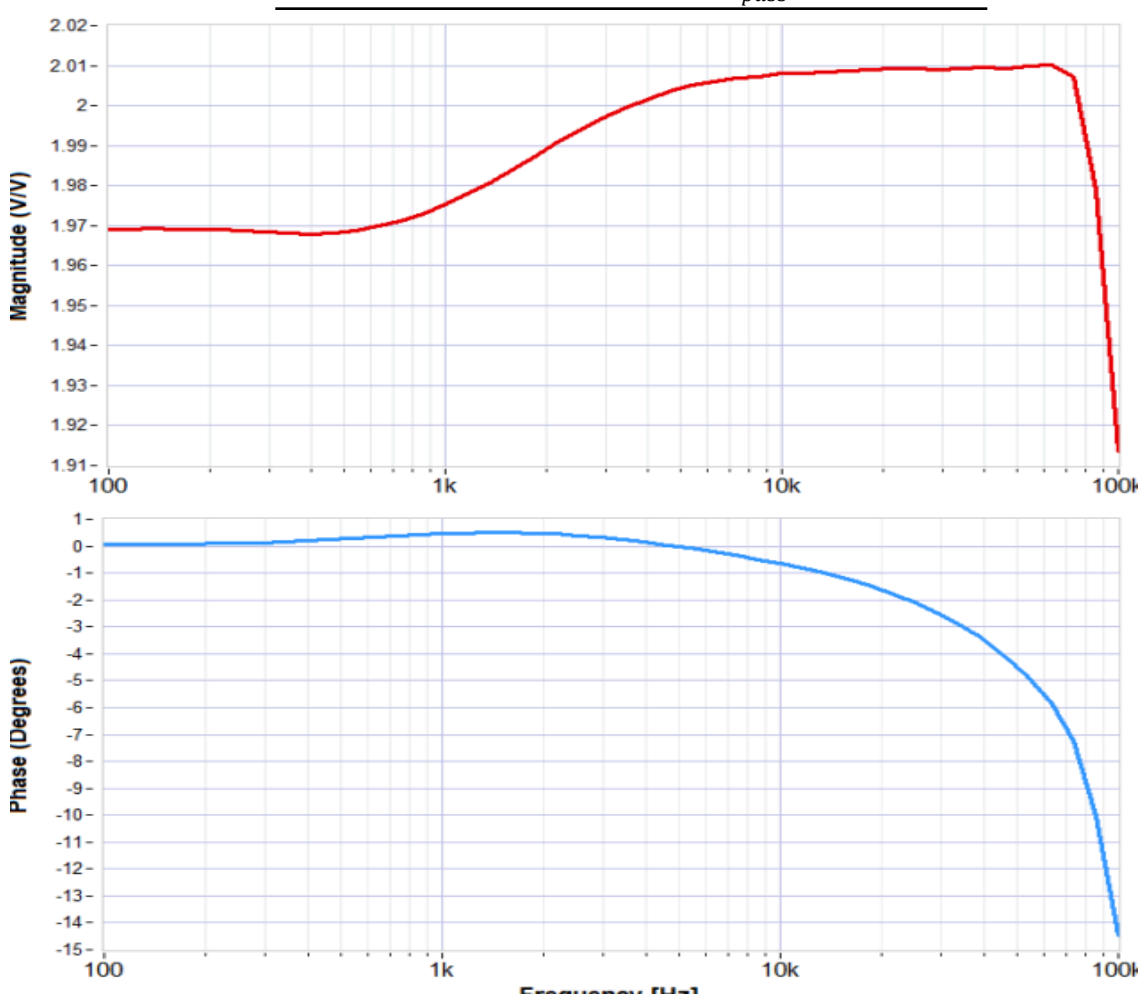


Figure 1: Bode Plot (Magnitude and Phase) of Measured Vs. Calculated Frequency Response

Conclusion

We picked standard RC values for passive low/high pass frequencies and used the LMC660 for the stages. We aimed for different output voltages given a set input voltage and variable frequencies. We first did it the “quick-and-dirty” way by sweeping the frequency from low to high to get data values. We also tried the same thing with the loopinator to create Bode plots. The hardest part was trying to read the oscilloscope when the frequency was very low, like at 10Hz. We couldn’t use the measure functions on the oscilloscope because there was too much noise. We had to use the cursors function. Some of the calculations were also pretty difficult to do. But all in all, the results on our graphs were very similar to what we were supposed to get, so we are happy about the results of this lab.